

ESP32 -HARNESS

ADVANCED PENTESTING, AUDIT & TELEMETRY FIRMWARE

v1.0 MIT

CONTROLLER

ESP32-WROOM-32

Dual-core 240 MHz, 520 KB SRAM

RF TRANSCEIVER

NRF24L01+ PA/LNA

2.4 GHz ISM, +20 dBm, 250 kbps - 2 Mbps

STORAGE

MicroSD Module

FAT32/exFAT, SPI, up to 32 GB

DISPLAY

OLED SSD1306

128x64 monochrome, I2C

BATTERY

LiPo 3.7V 1000mAh

USB-C charging, ~4 hours runtime

CAPABILITIES

Wi-Fi Research

802.11 frame capture, deauth monitoring, probe tracking

RF24 Spectrum

ISM band analysis, channel mapping, signal visualization

Packet Injection

Controlled RF transmission for protocol testing

BLE Enumeration

Bluetooth Low Energy device discovery

Forensic Capture

Full PCAP to SD, UTC timestamps, Wireshark export

GPIO Events

Hardware sensors, MQTT/HTTP telemetry

2.4
GHz

SYSTEM ARCHITECTURE

ESP32-WROOM-32
Main controller

Core Framework
CLI, PCAP Engine, Logger, Config

Wi-Fi Engine
Scan, Capture, Deauth detect

RF24 Engine
NRF24L01+ spectrum & injection

BLE Engine
Device enumeration & scanning

Telemetry Output
MQTT, HTTP, Serial

USAGE FLOW

- 01
- Flash firmware to device
- 02
- Load config (Wi-Fi, BLE)
- 03
- Select target network
- 04
- Capture packets (Wi-Fi)
- 05
- Export PCAP for analysis

COMPONENT MATRIX

COMPONENT	INTERFACE	PROTOCOL
ESP32-WROOM-32	UART / USB	Serial 115200
NRF24L01+ PA/LNA	SPI	SPI 10 MHz
MicroSD Module	SPI	SPI 25 MHz
OLED SSD1306	I2C	I2C 400 kHz
LiPo Battery	ADC	3.7V nominal